

Introduction to Information Retrieval

Bag of Visual Features based Image Retrieval

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Disclaimer

- Disclaimer: This curriculum have been inspired and adapted from materials created by:
 - Richard Szeliski, “Computer Vision: Algorithms and Applications”, 2nd edition, 2022. <https://szeliski.org/Book/>
 - Fei-Fei Li, “Computer Vision: Foundations and Applications” Course, Stanford University

Image Recognition

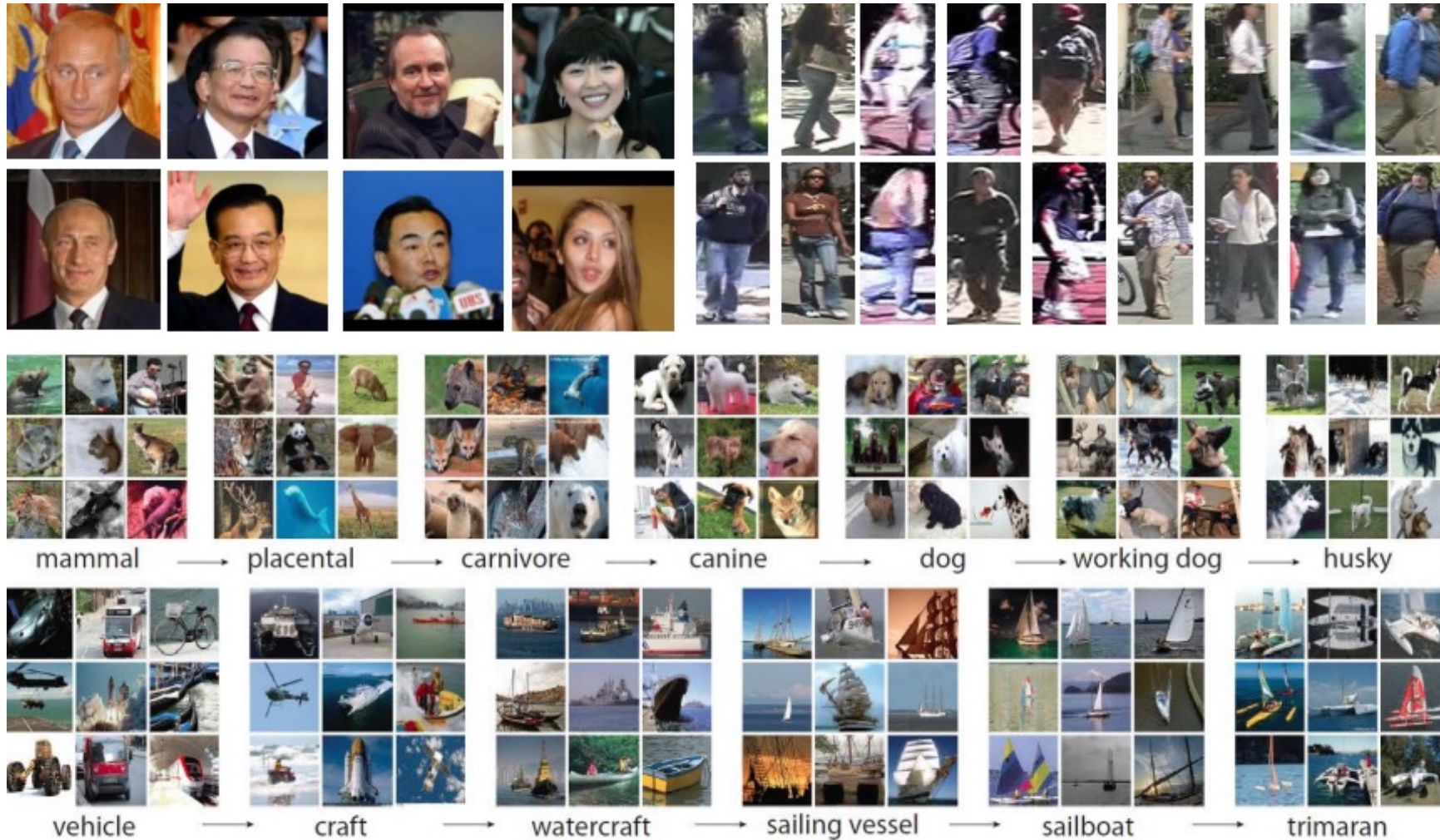


Image Retrieval



Precision: 90.00%



Precision: 72.22%



Precision: 55.56%



Precision: 44.44%



Precision: 94.44%



Precision: 66.67%



Precision: 55.56%



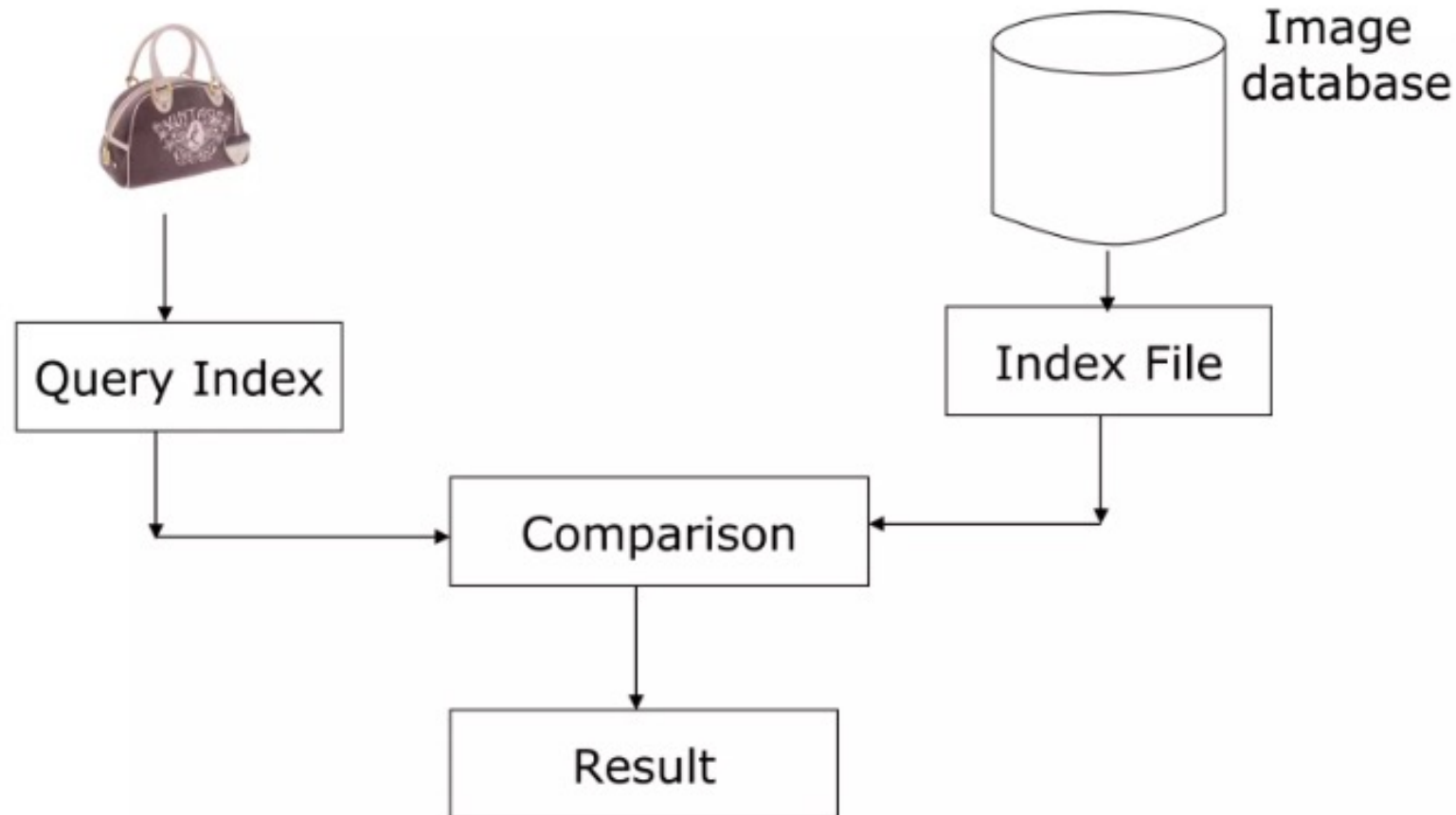
Precision: 36.11%

Image Retrieval System



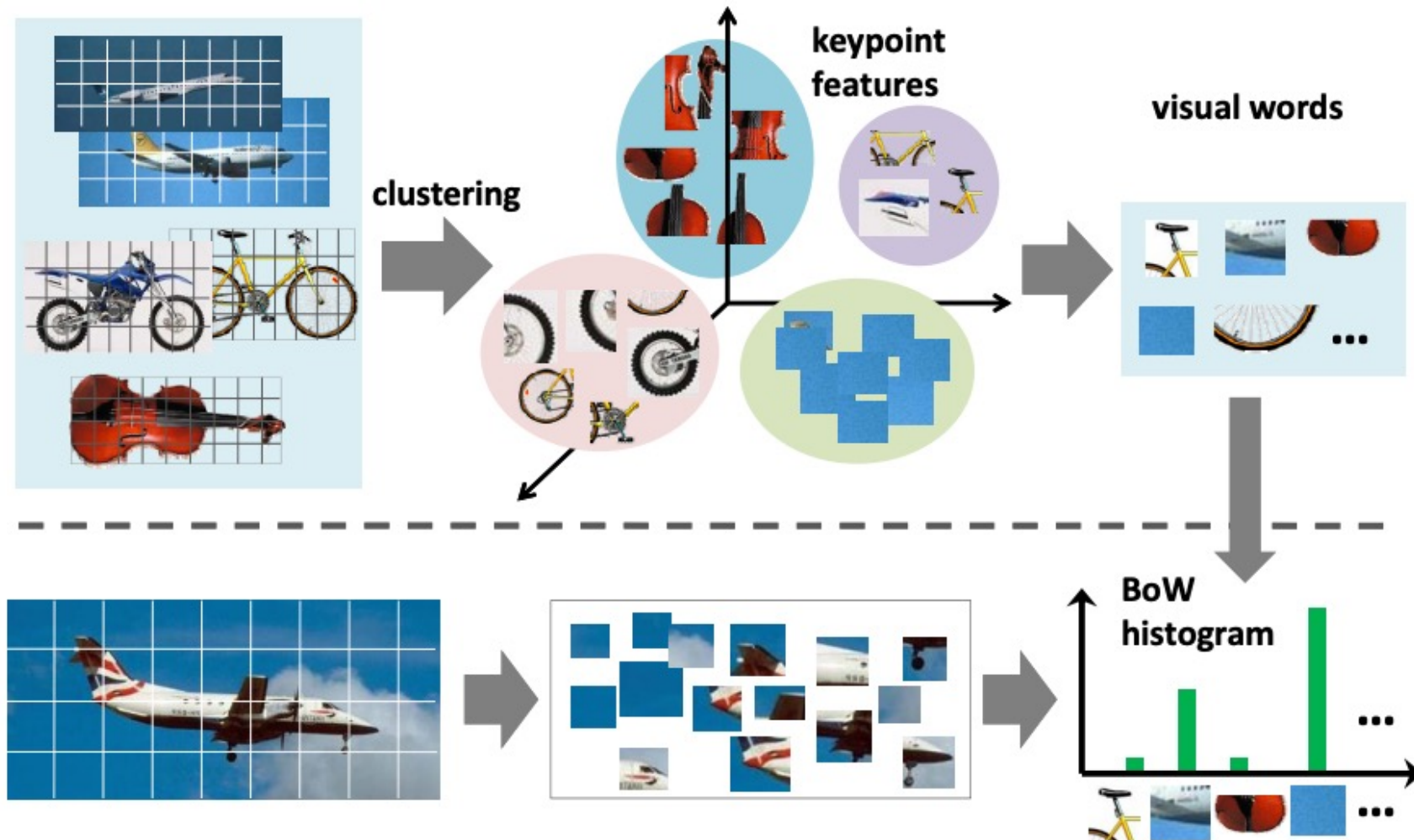


Visual Search a.k.a. Content Based Image Retrieval



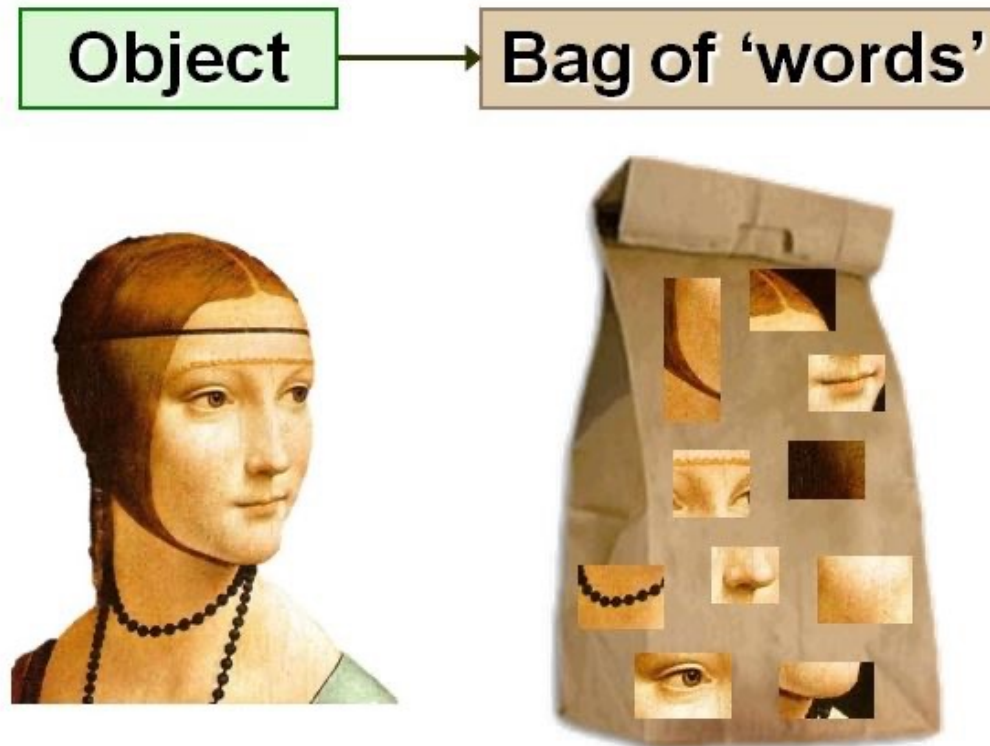
Block-diagram of a typical content-based image retrieval system

BoF-based Visual Search



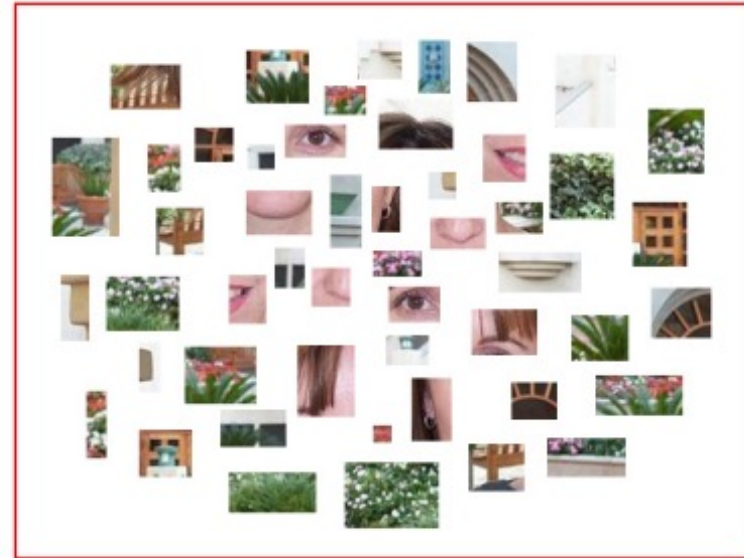
Bag of Features

- Bag of Features (BoF) = Bag of Visual Words (BoVW)
- Represent an image as a collection of unordered image patches



Visual Word

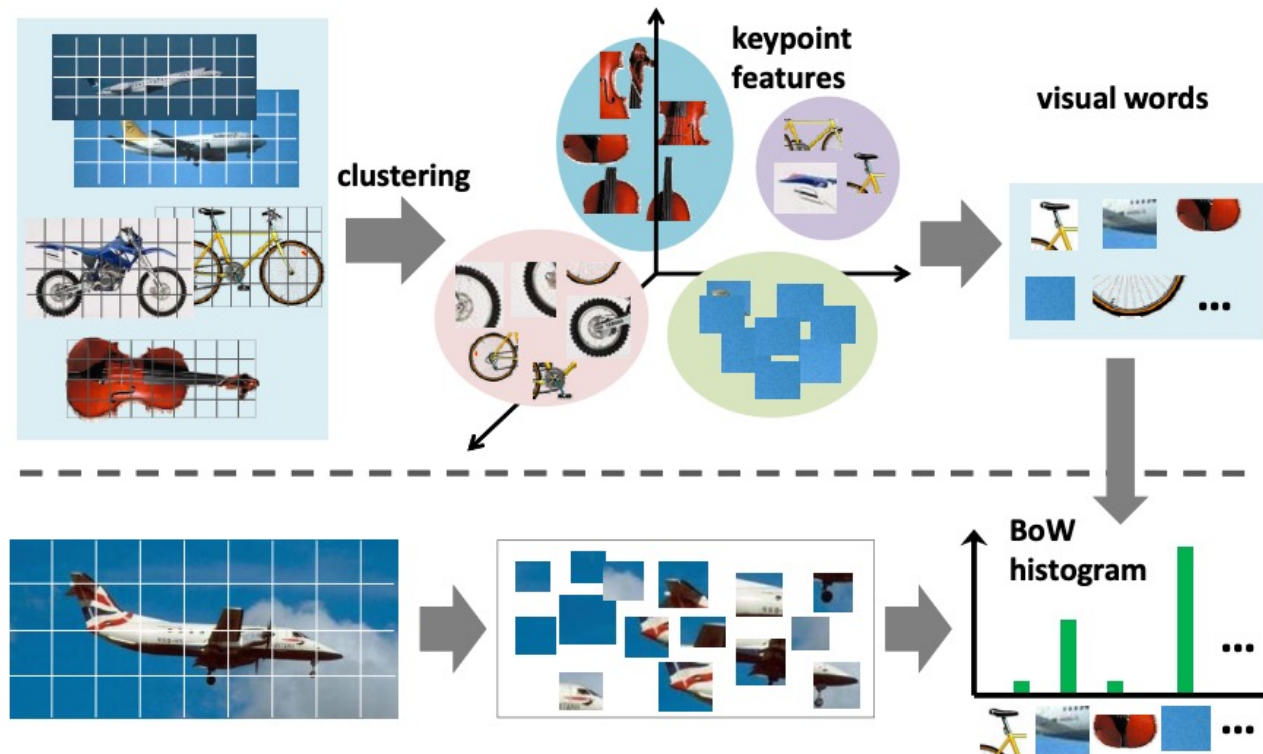
- Not randomly extract patches from image
- Visual words expectedly have meaning somehow



face, flowers, building

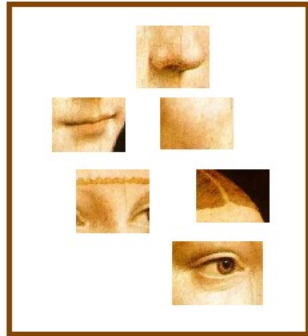
Idea of Bag of Features

- First, take a bunch of images, extract features, and build up a “dictionary” or “visual vocabulary” – a list of common features
- Given a new image, extract features and build a histogram – for each feature, find the closest visual word in the dictionary

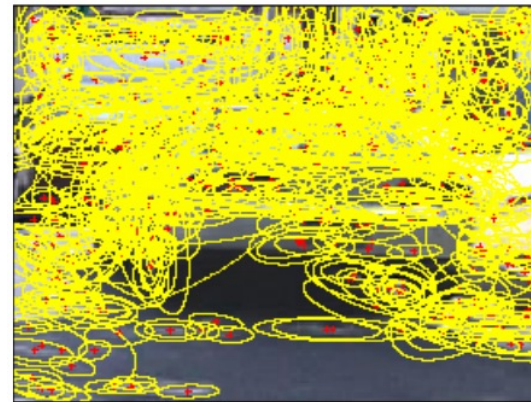


Bag of Features

1) Feature extraction:



- Extract local features: SIFT, SURF, ...

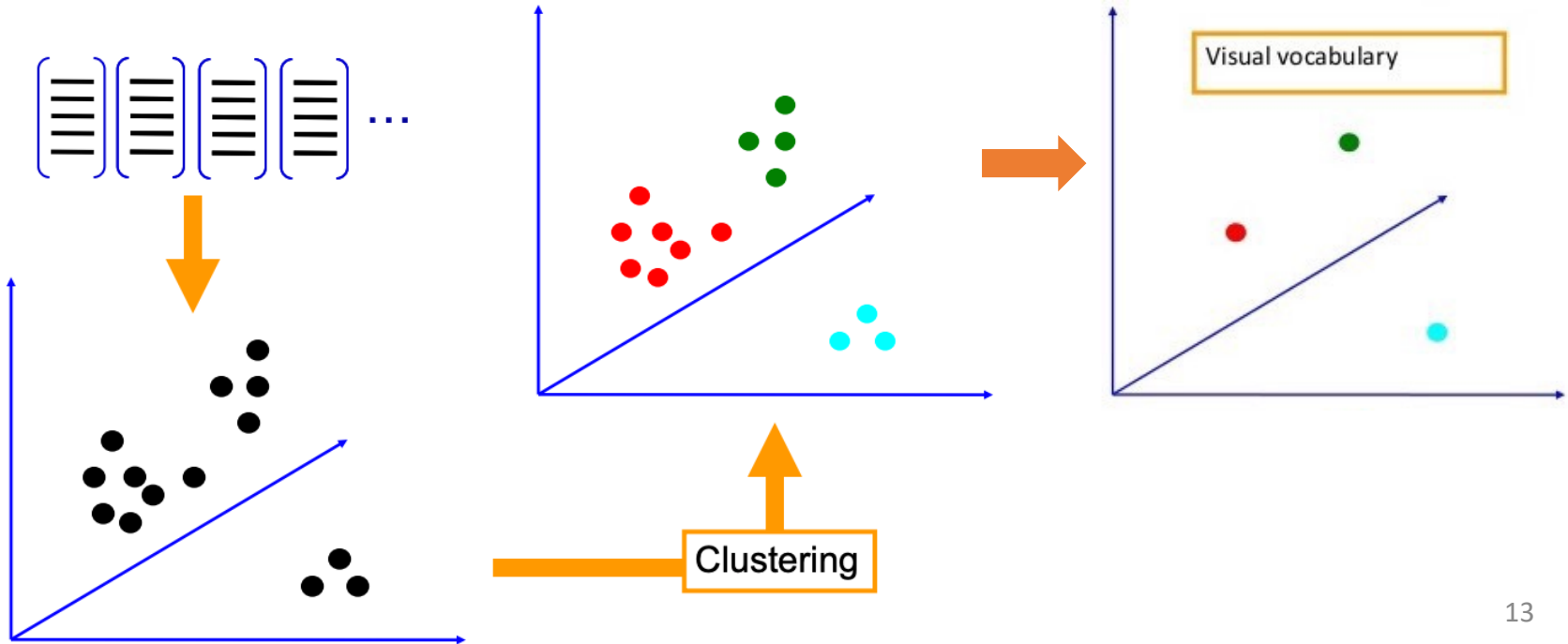


Bag of Features

2) Learn Visual Vocabulary



- Cluster local features (using K-means)



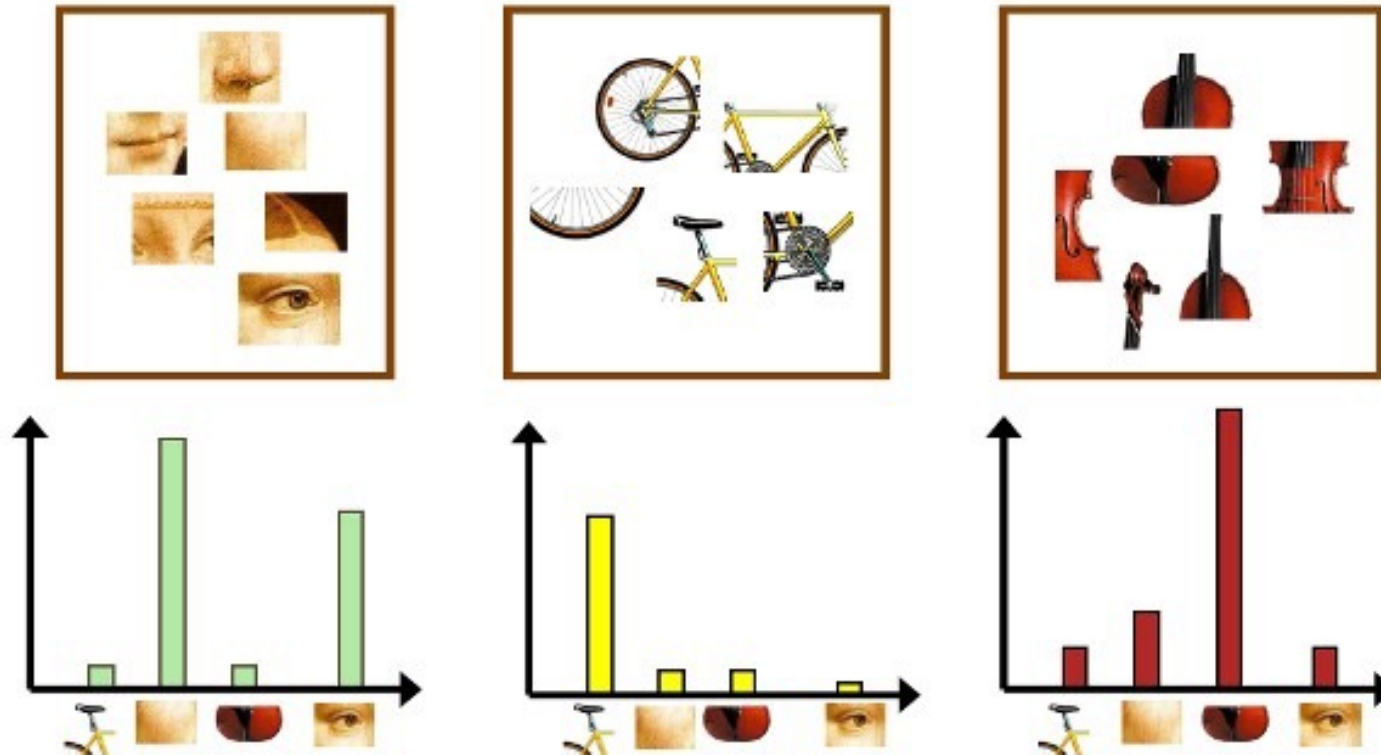
Bag of Features

3) Quantize features using visual vocabulary

- Codebook is used for quantizing features:
 - A vector quantizer takes a feature vector and maps it to the index of the nearest codevector in a codebook
 - Codebook = visual vocabulary
 - Codevector = visual word

Bag of Features

4) Represent images by frequencies of visual words



TF-IDF Weighting

- Instead of computing a regular histogram distance, we'll weight each word by its inverse document frequency

- Term frequency (TF) of word j in image I :

$$TF(j, I) = \frac{\text{number of occurrences of } j \text{ in image } I}{\text{number of words in image } I}$$

- Inverse document frequency (IDF) of word j :

$$IDF(j) = 1 + \log \frac{\text{number of documents}}{\text{number of documents in which } j \text{ appears}}$$

- To compute the value TF-IDF of bin j in image I :

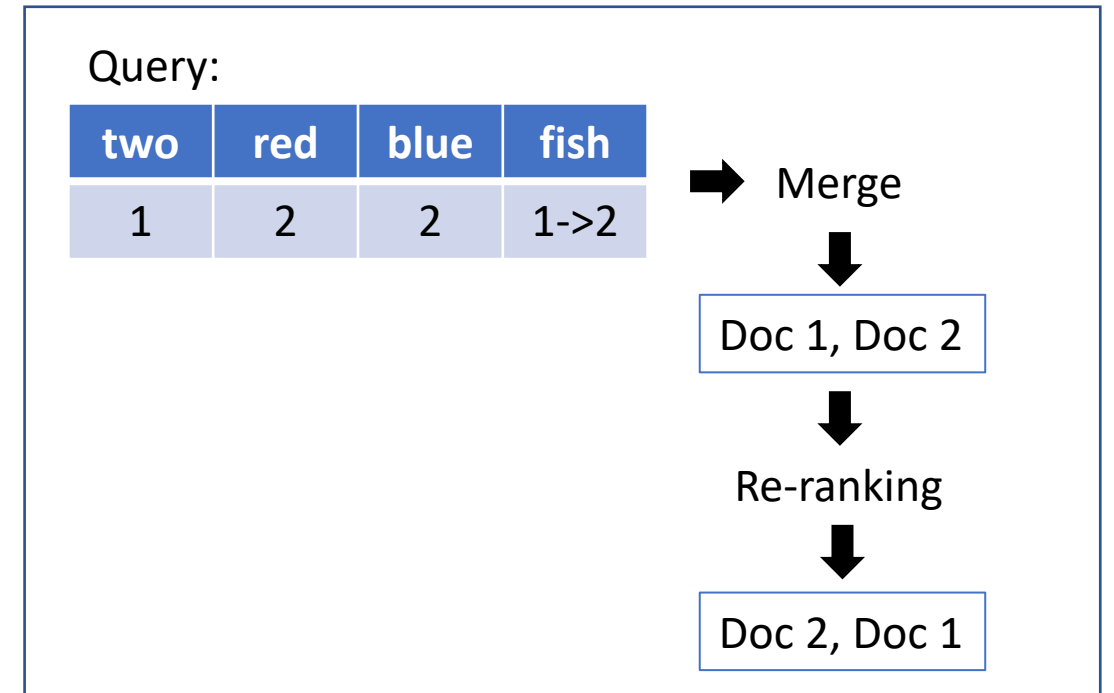
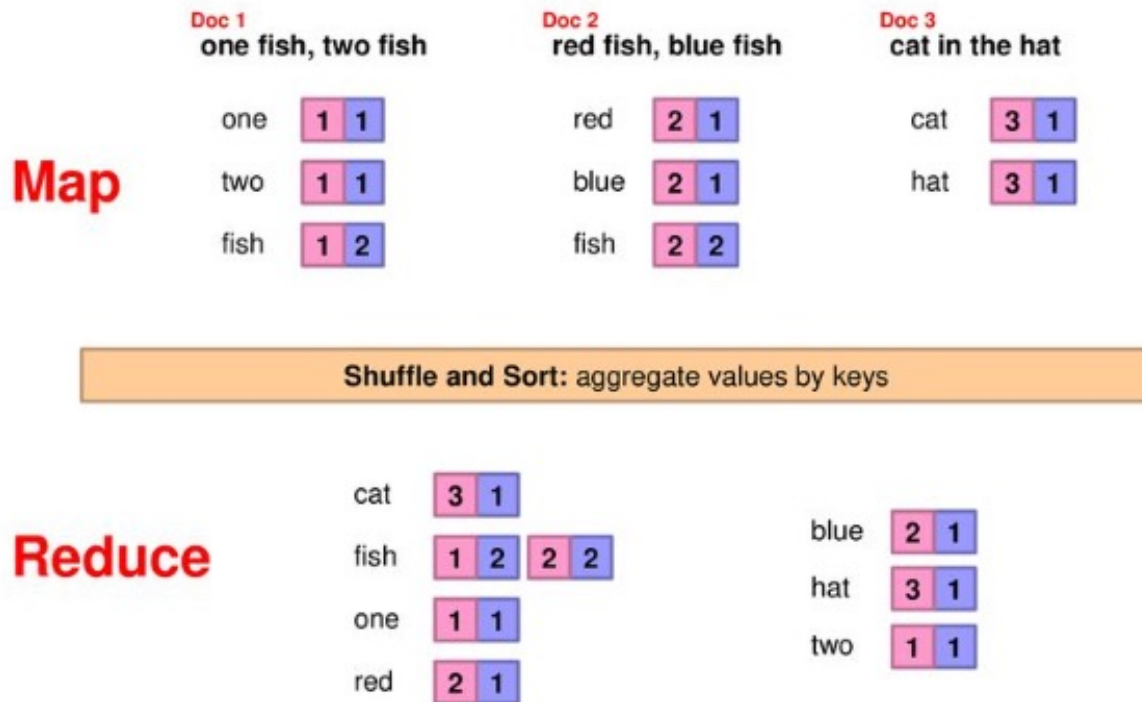
$$TFIDF(j, I) = TF(j, I) * IDF(j)$$

Inverted File (Inverted Index)

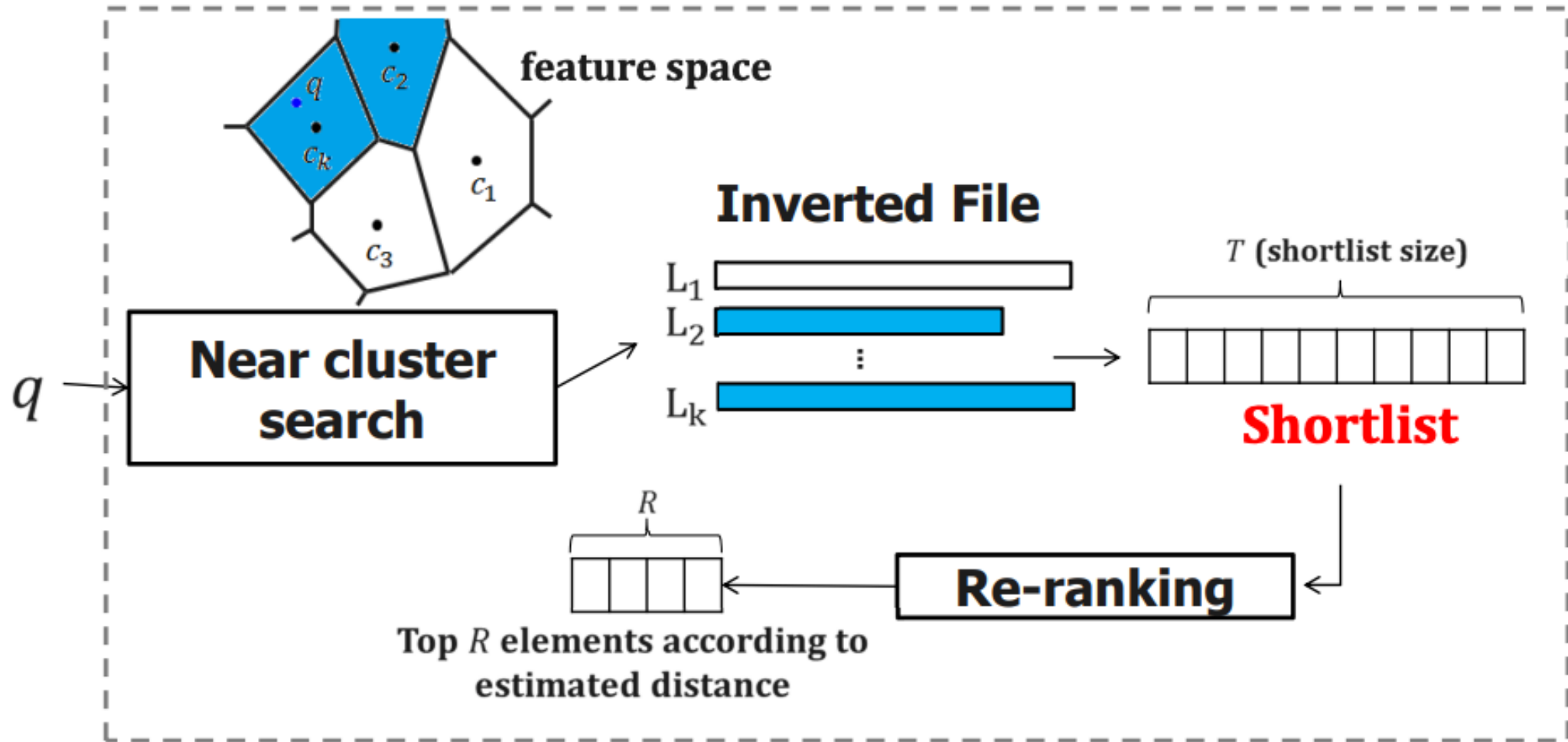
- Each image has $\sim 1,000$ features
- We have $\sim 100,000$ visual words
 - Each histogram is extremely sparse (mostly zeros)
- Inverted file
 - Mapping from words to documents
- Can quickly use the inverted file to compute similarity between a new image and all the images in the database
 - Only consider database images whose bins overlap the query image

Inverted File (Inverted Index)

- For each word, list images containing the word



Inverted File (Inverted Index)



Inverted File (Inverted Index)

Database images



Image #1



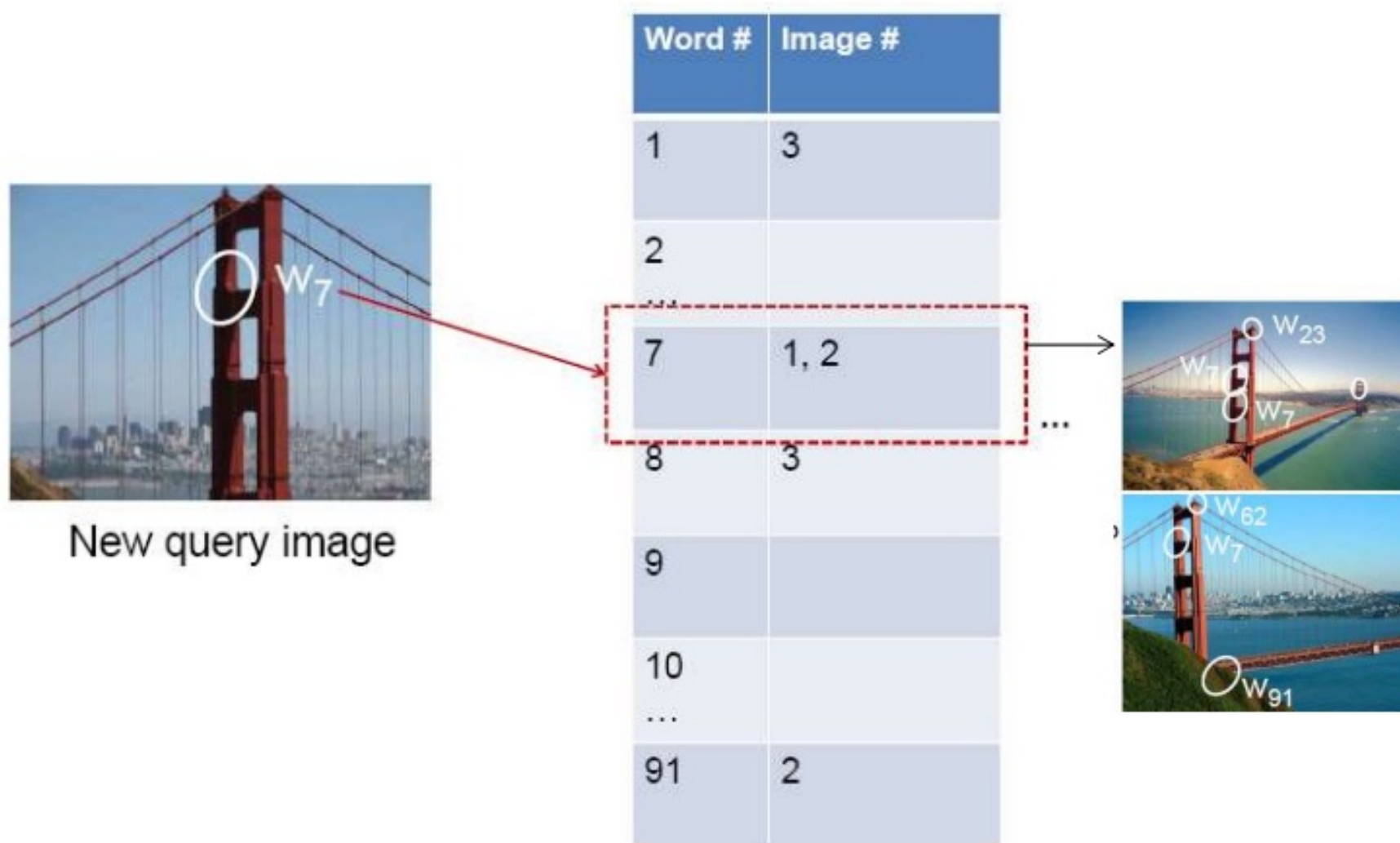
Image #2



Image #3

Word #	Image #
1	3
2	
...	
7	1, 2
8	3
9	
10	
...	
91	2

Inverted File (Inverted Index)



Large-Scale Image Search

- Build the database:
 - Extract features from the database images
 - Learn a vocabulary using k-means (typical k: 100,000)
 - Compute weights for each word
 - Create an inverted file mapping words to images

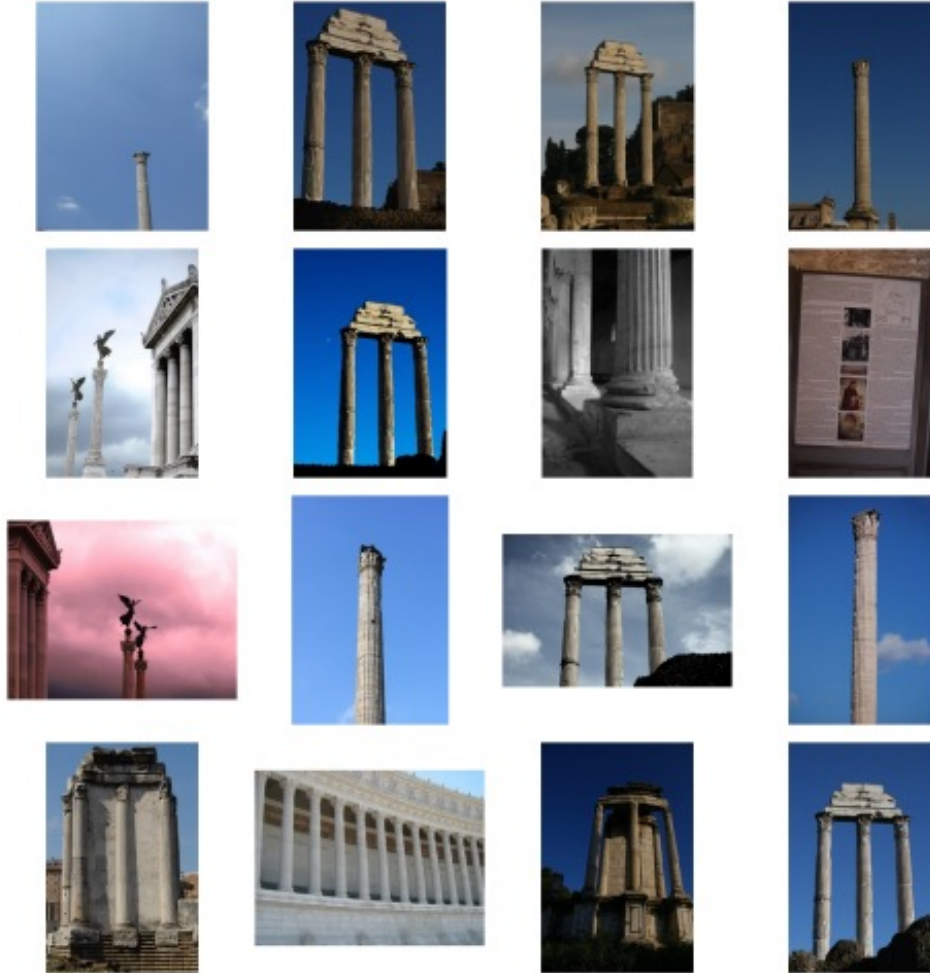


Large-Scale Image Search

- Cons:
 - Performance degrades as the database grows
- Pros:
 - Works well for CD covers, movie posters
 - Real-time performance possible (database of 40K CD covers [Nister et al., 2006])

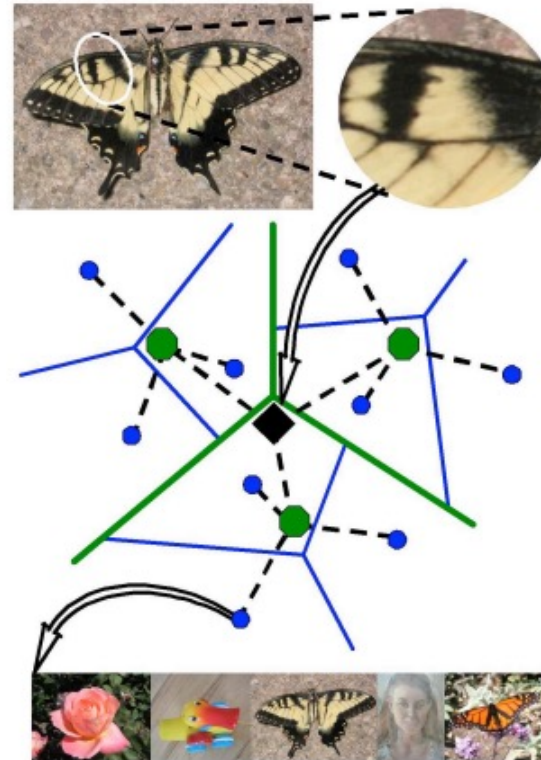


Example of BOF

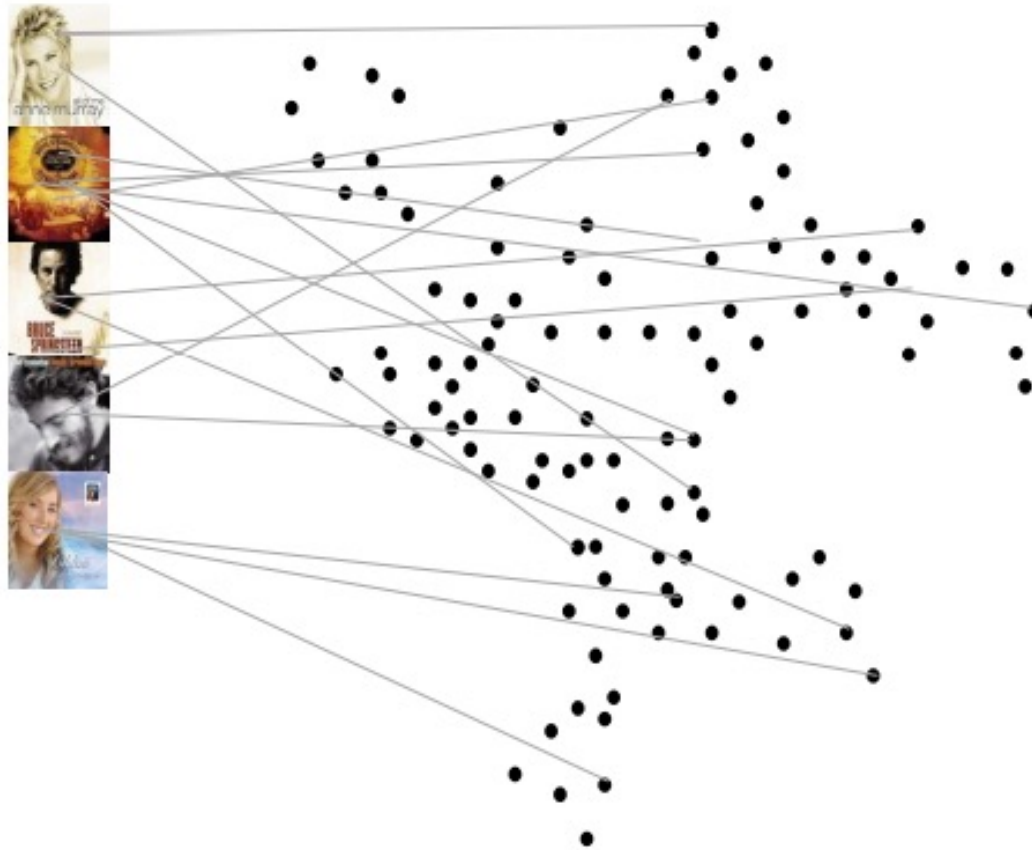


Issues of Visual Vocabularies

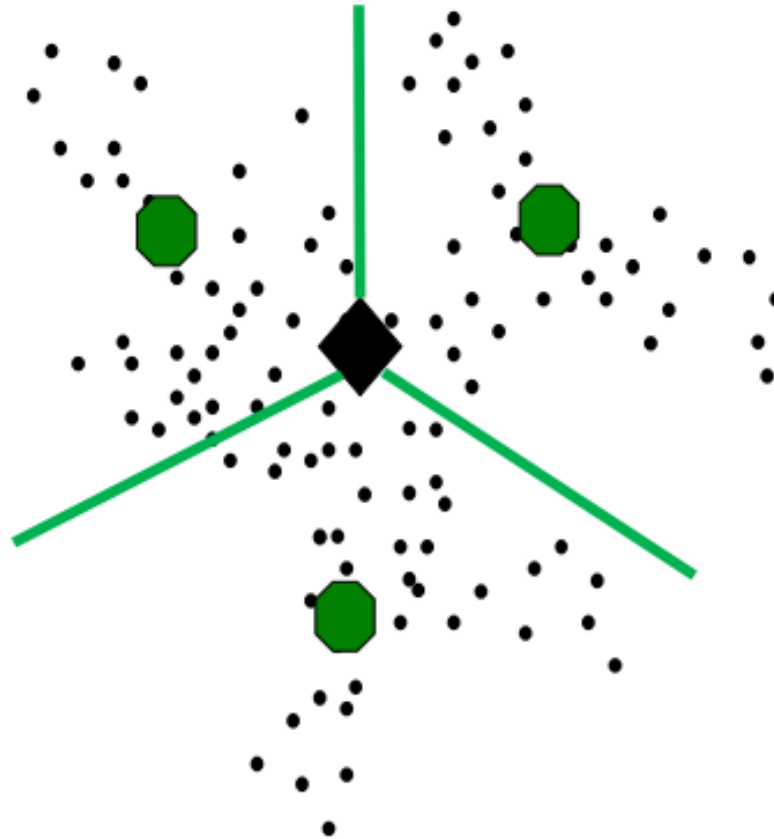
- How to choose vocabulary size?
 - Too small: visual words not representative of all patches
 - Too large: quantization artifacts, overfitting
- Computational efficiency
 - Vocabulary trees (Nister et al., 2006)
(hierarchical k-means clustering)



Growing Vocabulary Tree

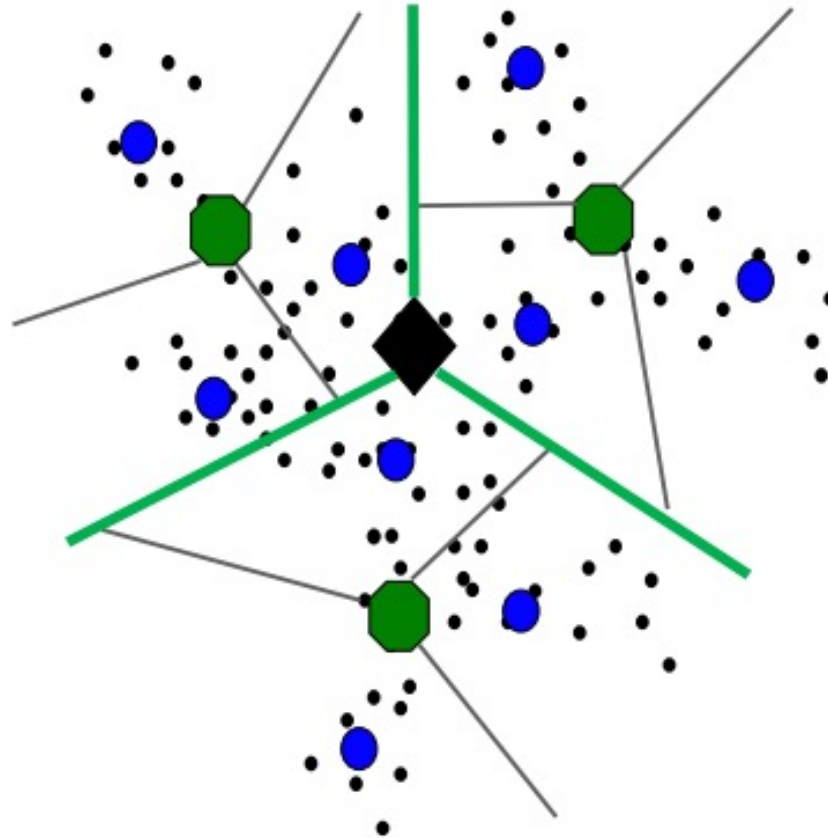


Growing Vocabulary Tree



$k = 3$

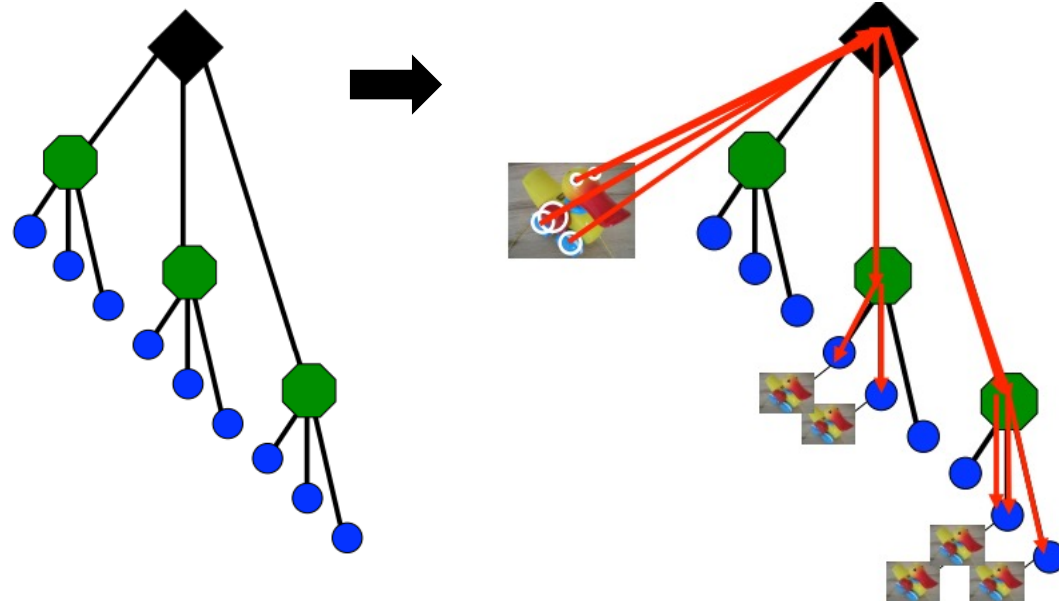
Growing Vocabulary Tree



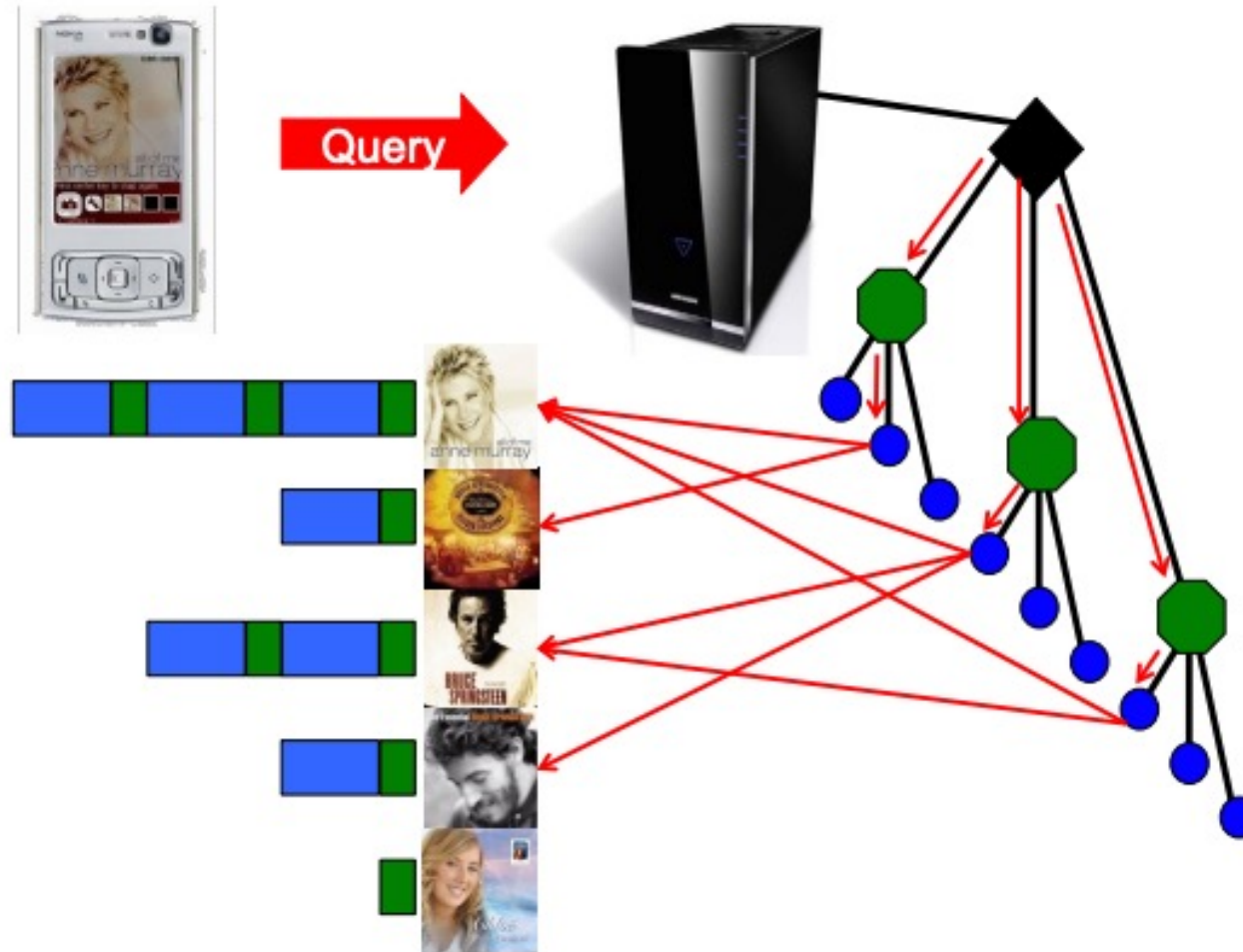
$k = 3$

Vocabulary Tree

- As vocabulary tree is formed and is online, new images can be inserted in database
- Adding image to the database requires following steps
 - Image feature descriptors are computed
 - Each descriptor vector is dropped down from the root of the tree and quantized into a path down the tree

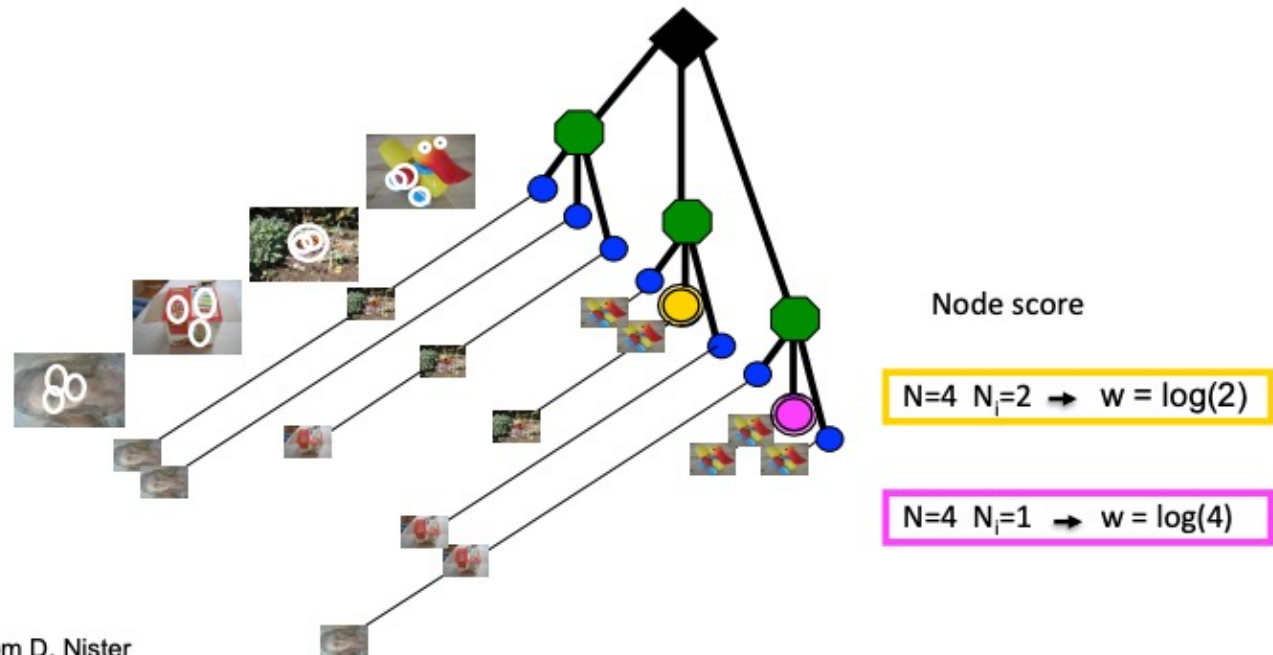


Querying Vocabulary Tree



Scoring

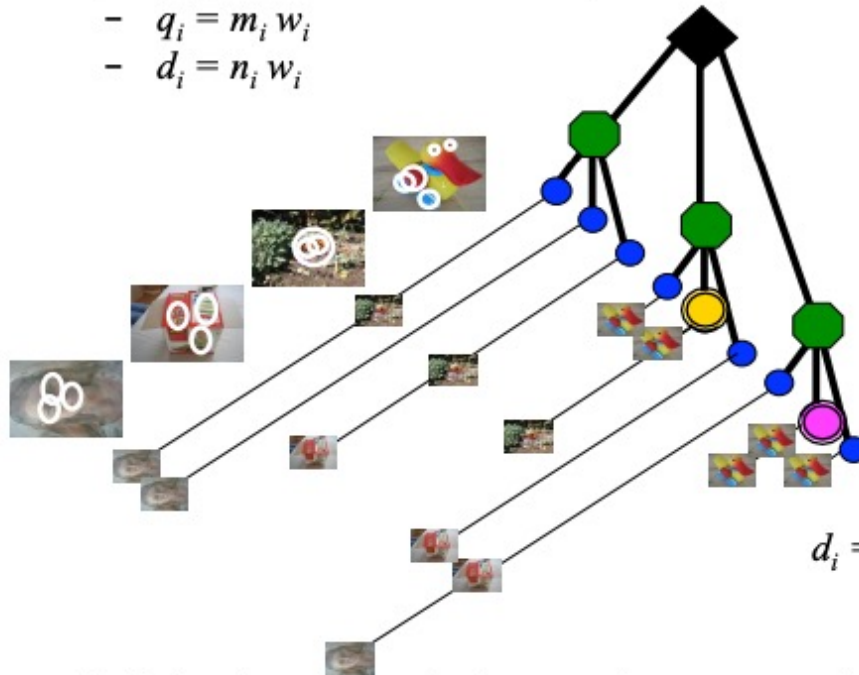
- At each node i a weight w_i is assigned that can be defined according to one of different schemes:
 - Constant weighing scheme: $w_i = k$
 - Entropy weighing scheme: $w_i = \log\left(\frac{N}{N_i}\right)$ (IDF)



Weighted Vocabulary Tree

- Query q_i and database vectors d_i are defined according to the assigned weights as :

- $q_i = m_i w_i$
- $d_i = n_i w_i$



where m_i is the number of the descriptor vectors of the query with a path along the node i and w_i its weight

n_i is the the number of the descriptor vectors of each database image with a path along the node i .

$$d_i = n_i w_i$$

$$S = 2 \log(2)$$

$$S = 2 \log(4)$$

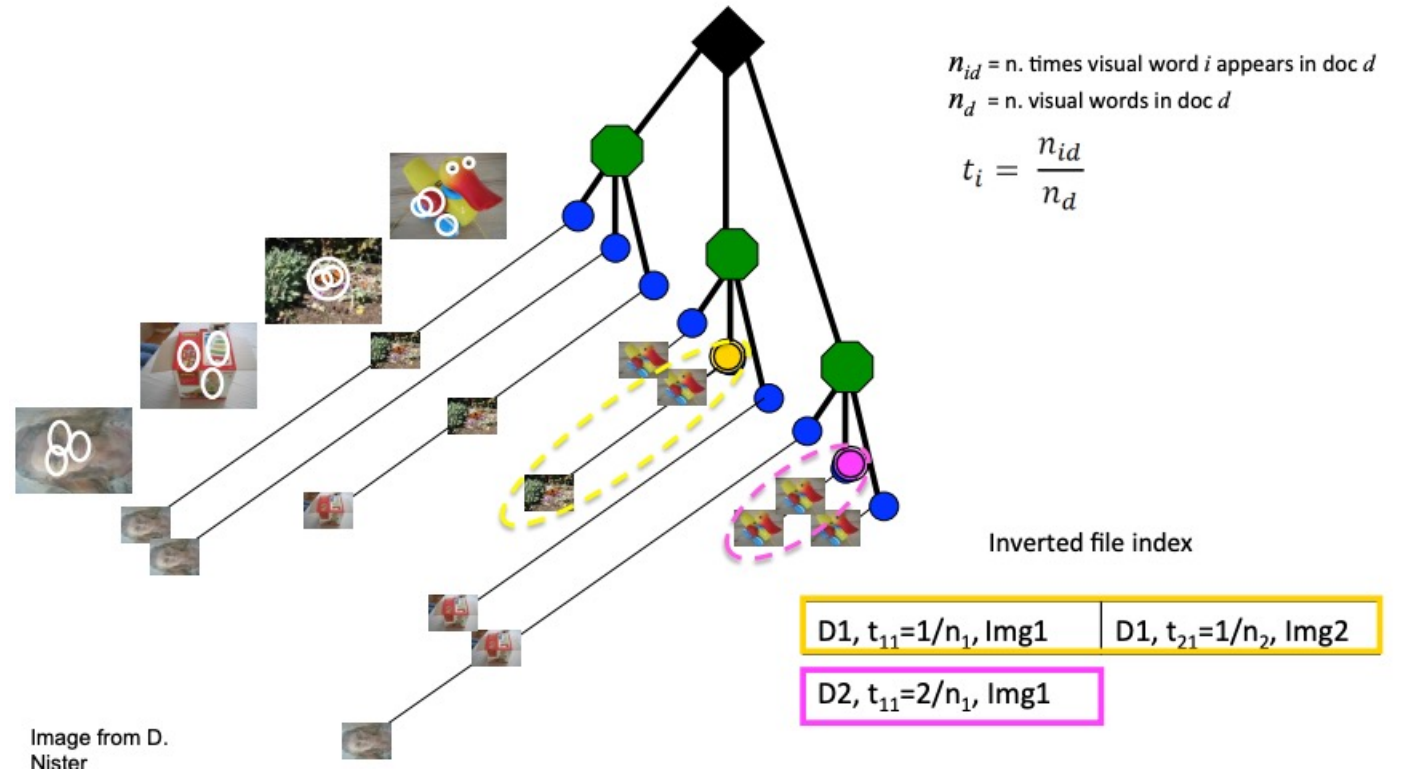
- Each database image is given a relevance score based on the L1 normalized difference between the query and the database vectors

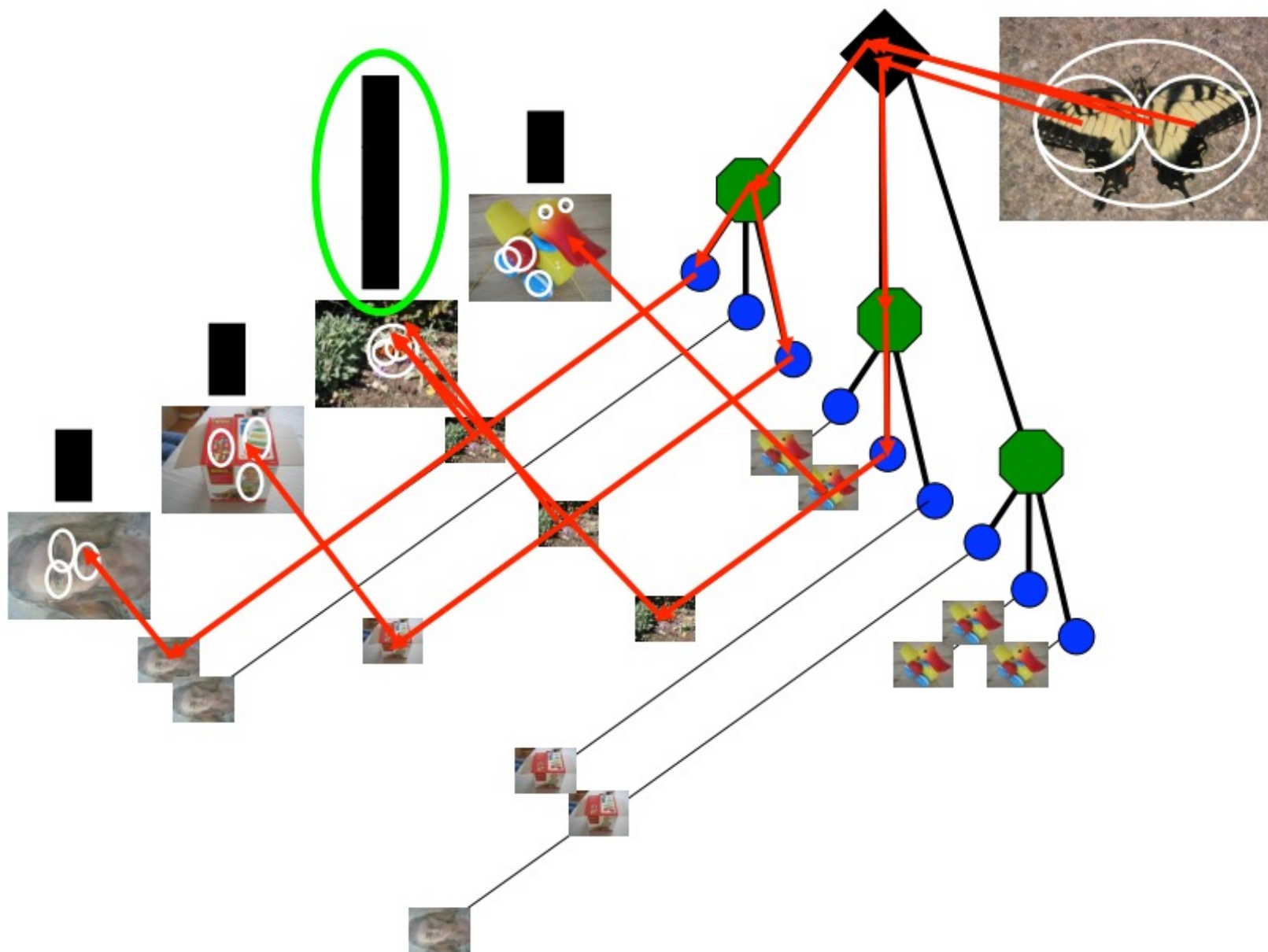
$$s(q, d) = \left\| \frac{q}{\|q\|} - \frac{d}{\|d\|} \right\|$$

- Scores for the images in the database are accumulated. The winner is the image in the database with the most common information with the input image.

Embedded Inverted File in Vocabulary Tree

- Inverted file of inner nodes is the concatenation of its children's inverted files
- Inverted files at each node store id-numbers of images in which a particular node occurs and the TF of that image. Indices back to new image are then added to relevant inverted files.

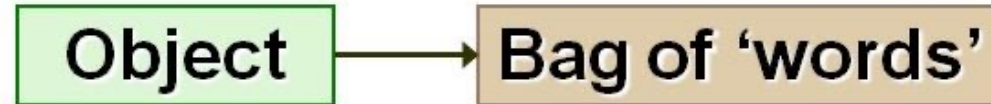




Slide from D. Nister

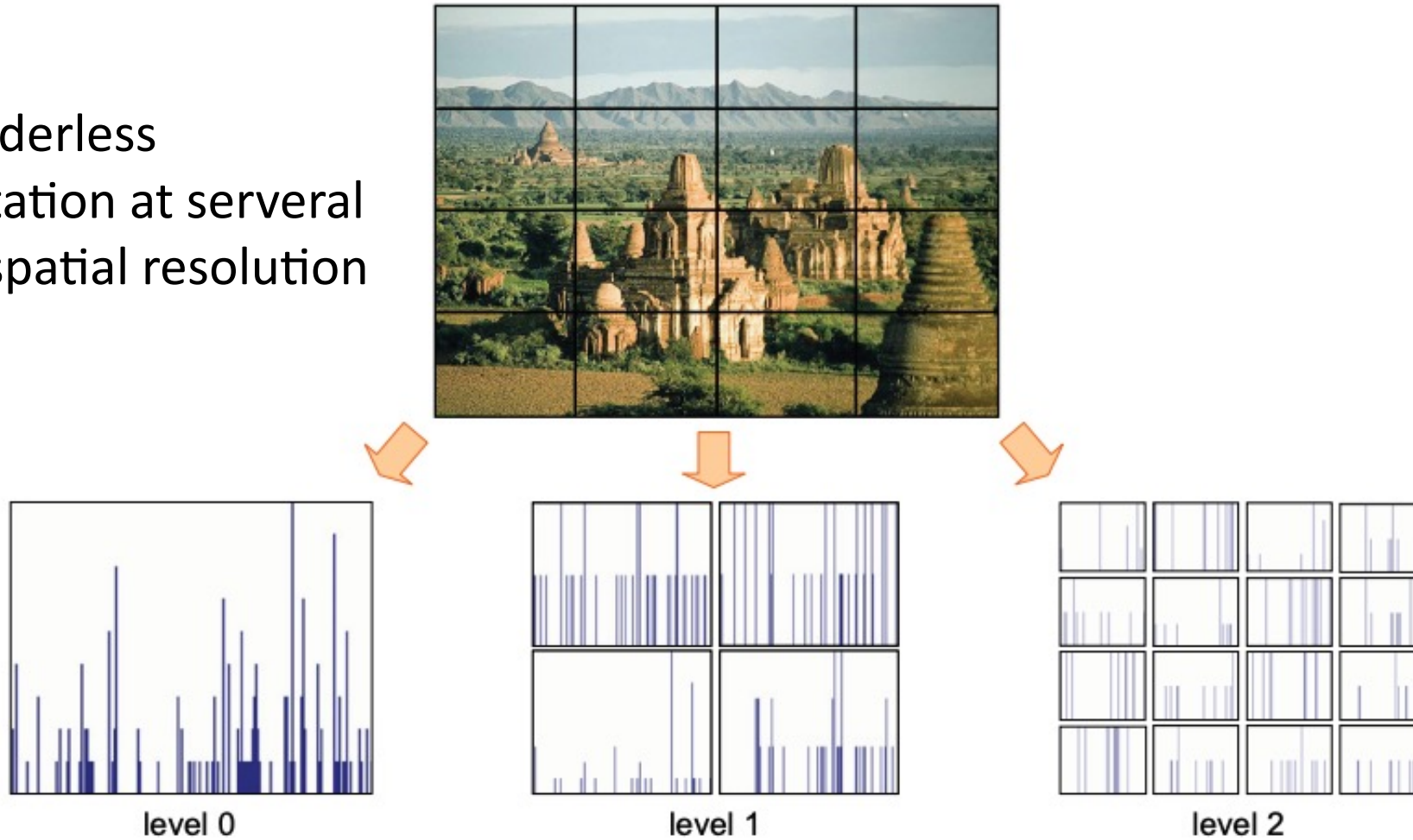
Issues of BoF

- Completely disregard spatial layout of features in image
- Cannot take advantage of regularities in image composition and spatial arrangement of features



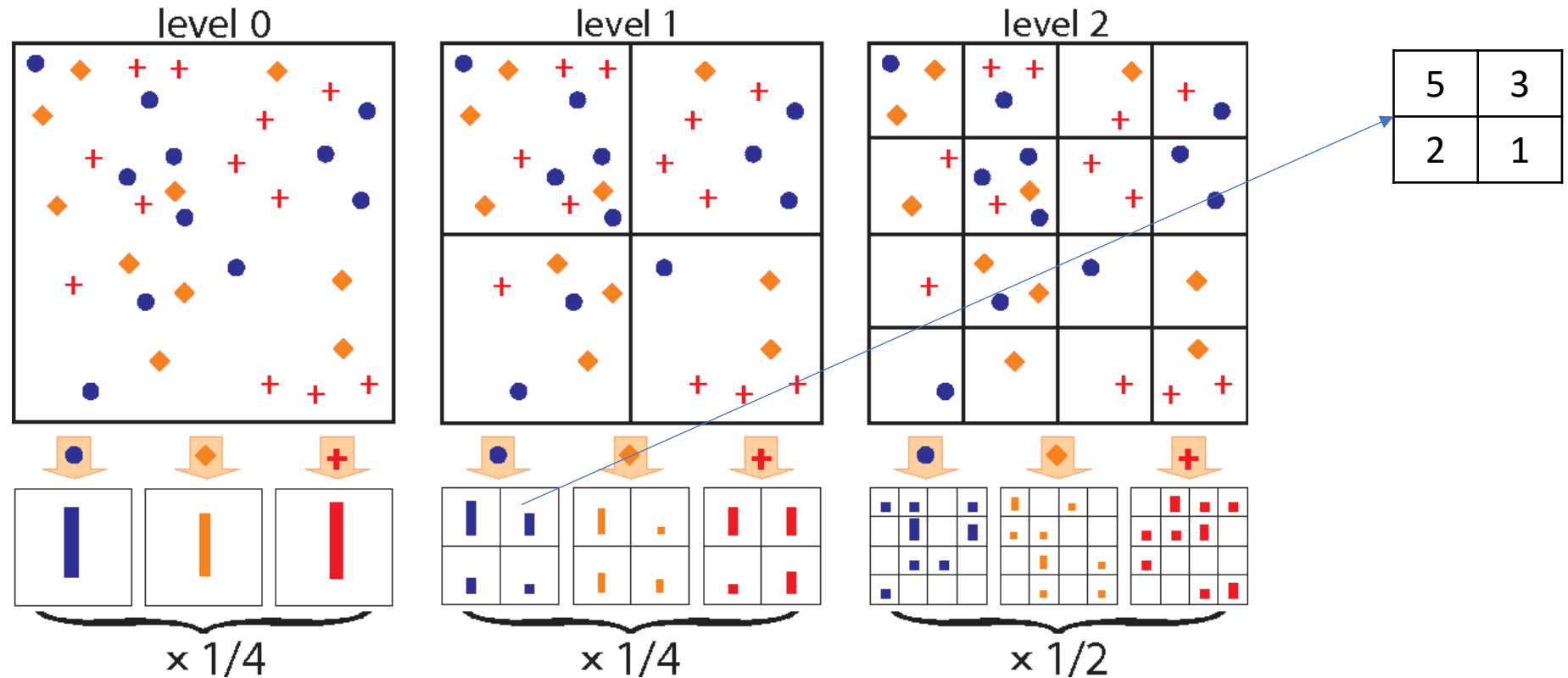
Spatial Pyramid Representation

Locally orderless
representation at several
levels of spatial resolution



Spatial Pyramid Matching (Lazebnik et al., 2006)

- Combination of spatial levels with pyramid match kernel



Spatial Pyramid Matching

